

BOUGOUNI, Mali

WMO: 612960

Lat: 11.417N

Lon: 7.500W

Elev: 1152

StdP: 14.09

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	61.8	63.8	31.3	26.6	79.2	34.3	30.2	78.0	10.0	78.2	9.3	80.1	1.2	N/A	0.305

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	21.3	103.7	73.6	102.2	73.4	100.6	73.5	82.1	92.7	81.2	91.7	80.6	90.8	3.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.5	160.0	88.2	78.7	155.4	87.0	78.0	151.8	86.2	46.9	93.0	46.0	92.2	45.2	90.7	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.5	6.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.8	78.5	83.8	89.1	90.0	87.0	83.4	80.0	79.4	80.2	82.5	81.9	78.5
	DBStd	4.91	3.34	4.06	3.38	3.11	3.76	3.45	2.59	2.30	2.17	2.40	2.42	3.22
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11990	885	946	1213	1201	1146	1003	931	910	907	1008	956	884
	CDD65	6515	420	526	748	751	681	553	466	445	457	543	506	419
	CDH74	76506	4555	6611	10868	11219	9155	6491	4258	3607	3912	5636	5656	4538
	CDH80	40103	2349	3800	6825	7073	5172	3069	1461	1052	1385	2529	3051	2337
Wind	WSAvg	1.7	2.1	2.4	2.1	2.5	2.1	1.8	1.0	1.1	1.1	1.1	1.4	2.2
Precipitation	PrecAvg	44.6	0.0	0.0	0.3	1.7	4.1	6.0	10.2	10.8	8.3	2.9	0.3	0.0
	PrecMax	58.6	0.5	0.5	1.7	6.8	11.5	10.0	16.0	23.3	15.4	8.9	1.9	0.8
	PrecMin	33.2	0.0	0.0	0.0	0.0	0.3	2.7	5.6	4.7	3.9	0.1	0.0	0.0
	PrecStd	6.8	0.1	0.1	0.4	1.5	2.2	2.0	2.7	3.3	3.0	2.1	0.5	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.7	103.1	105.5	105.9	103.9	99.5	92.5	90.7	92.4	96.6	98.7	98.0
		MCWB	67.9	75.6	71.5	73.4	77.3	78.9	79.4	80.5	81.1	74.9	71.3	68.4
	2%	DB	95.3	100.7	103.5	103.9	101.0	96.9	90.4	88.8	90.4	94.4	96.8	95.7
		MCWB	66.6	71.3	71.3	74.4	78.3	79.5	79.3	79.7	80.6	77.5	71.0	67.7
	5%	DB	93.4	98.6	102.1	102.4	98.8	94.3	88.7	87.2	88.9	92.6	95.3	93.6
		MCWB	65.6	69.7	71.3	74.3	78.6	79.5	79.0	79.1	80.0	78.4	71.3	66.7
	10%	DB	91.4	96.5	100.6	100.5	96.7	92.1	87.0	85.7	87.1	90.8	93.6	91.3
		MCWB	64.8	68.3	71.6	74.6	78.6	79.3	78.5	78.4	79.3	78.5	71.6	65.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.9	81.8	83.1	80.9	83.1	82.7	81.1	81.8	83.0	81.8	78.9	74.2
		MCDB	91.4	101.1	95.4	98.2	96.1	93.1	89.0	88.5	90.2	90.8	91.4	89.8
	2%	WB	69.8	76.6	77.8	79.7	81.9	81.5	80.1	80.4	81.3	80.8	77.4	71.4
		MCDB	89.5	95.2	95.1	97.0	94.6	92.1	87.9	87.4	88.8	90.0	89.3	90.4
	5%	WB	67.8	73.9	76.5	78.9	80.9	80.6	79.4	79.5	80.4	80.0	76.2	69.0
		MCDB	89.4	91.8	95.5	96.0	93.5	91.1	87.0	85.9	87.7	89.1	87.8	89.4
	10%	WB	65.9	71.3	75.2	78.0	80.0	79.8	78.7	78.7	79.5	79.1	74.9	67.1
		MCDB	88.7	89.1	94.4	94.3	92.4	89.8	86.1	84.8	86.3	88.0	86.2	88.1
Mean Daily Temperature Range	5% DB	MDBR	27.3	25.7	24.2	21.3	19.0	16.6	13.3	12.3	14.6	17.9	26.1	28.1
		MCDBR	29.3	27.8	25.8	24.0	21.8	19.6	15.9	14.9	17.0	20.3	28.0	30.2
		MCWBR	12.0	11.8	9.8	7.2	6.8	7.0	6.8	7.1	8.6	7.4	8.8	11.1
	5% WB	MCDBR	27.5	24.2	21.6	20.5	19.6	17.7	15.2	14.3	16.5	18.0	21.4	29.0
		MCWBR	12.7	12.3	9.1	6.9	7.3	7.8	7.3	7.5	8.9	8.4	7.9	12.5
Clear-Sky Solar Irradiance	taub	0.527	0.591	0.725	0.679	0.621	0.582	0.523	0.491	0.503	0.538	0.500	0.483	
	taud	1.912	1.769	1.548	1.697	1.806	1.917	2.105	2.208	2.147	2.019	2.040	2.040	
	Ebn at Noon	247	236	209	218	226	232	248	259	259	247	253	256	
	Edh at Noon	61	73	92	79	69	61	51	47	50	56	53	52	
All-Sky Solar Radiation	RadAvg	1807	1924	1997	1968	1927	1807	1691	1618	1727	1843	1802	1758	
	RadStd	89	89	87	99	75	69	62	61	75	48	60	58	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	+0.54	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page